

How BloodLink works, and how to test all of it

A blood-donation platform built around registered **blood banks**. Three roles, one spine: a **member** asks for blood, the system scores and ranks compatible members by availability and distance, a hospital **officer** matches one and records the donation into their bank's stock, and everyone locates and routes to the nearest bank. This manual walks each role end to end, maps how the pieces connect, explains the enhancements that make it "enhanced," and gives you a role-by-role QA script.

01 / The cast

Three roles, at a glance

Everyone signs in through one of two doors — the admin portal (`login.php`) or the shared user portal (`user-login.php` , role dropdown: Member or Hospital Officer). The session `role` gates every page.

SESSION ROLE: ADMIN

Administrator

Runs the bank network: manages blood banks, members, hospital officers, and per-bank stock; posts emergency alerts; monitors the request queue; searches the donor pool; reads charts and reports.

SESSION ROLE: OFFICER

Hospital Officer

Belongs to a blood bank. Works the request queue and the Emergency Board, matches each request to the best-scored available member, and **confirms the donation** — which banks the units into their own bank's stock.

SESSION ROLE: MEMBER

Member

One account that both **donates** and **requests**. Registers with a blood group + address, sets availability (status, schedule, contact), checks eligibility (56-day rule), files blood requests, tracks their status, sees alerts, and routes to the nearest bank.

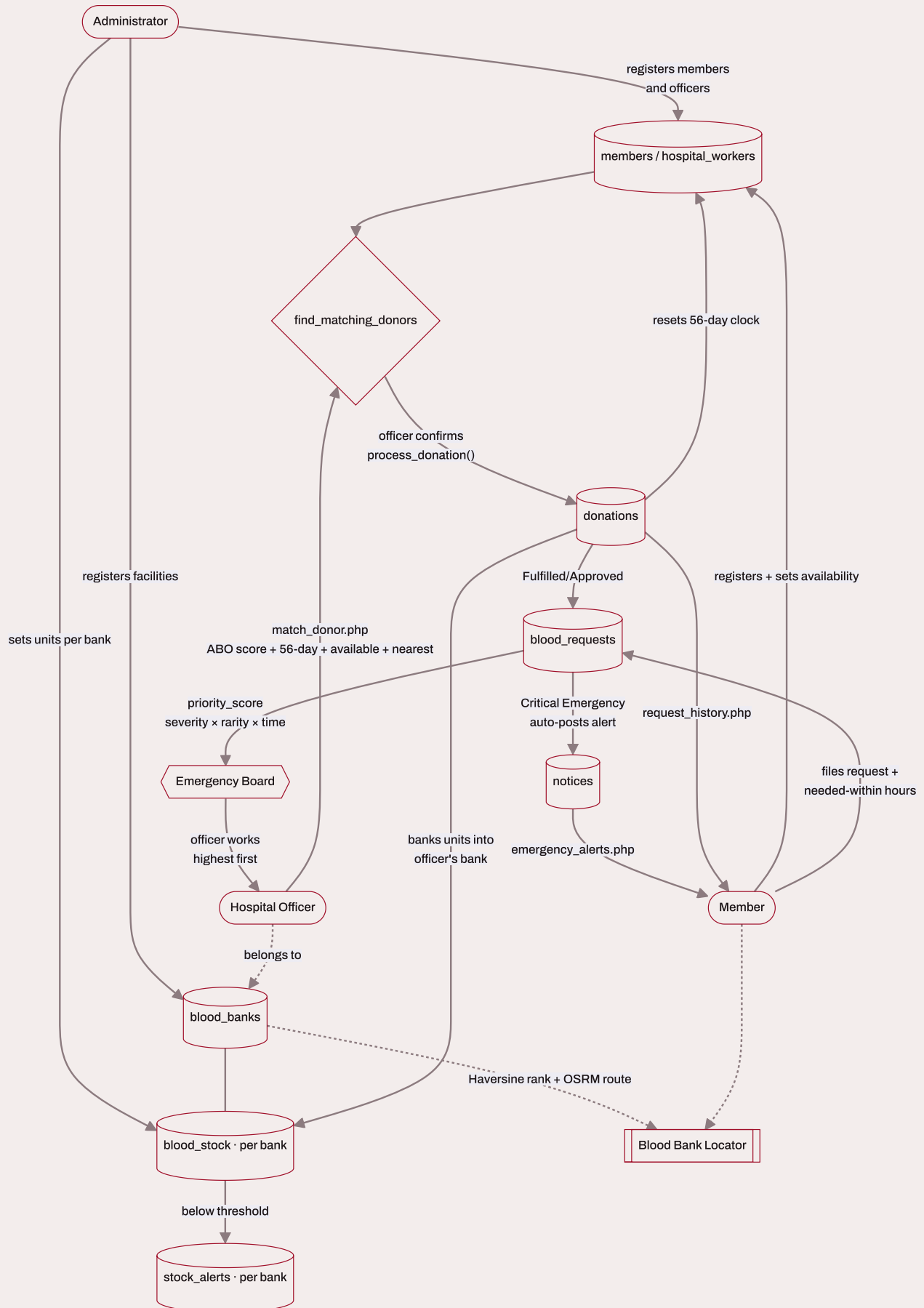
WHY A SINGLE "MEMBER" ACCOUNT

A real person can both give and need blood, so BloodLink uses **one member account for both** rather than separate donor and recipient logins — every member is a searchable donor *and* can file requests. On top of that base, four enhancements make it "enhanced": a **donor availability registry** with a Find Donors search, an **emergency workflow** that scores and ranks requests by priority with live countdowns, **compatibility-aware matching** (ABO/Rh score + blood-type rarity, availability-first), and **analytics dashboards**. Section 05 covers all four.

02 / The spine

How the whole app connects

Everything orbits one workflow: a request becomes a matched, recorded donation banked at a specific facility. The locator, the 56-day rule, availability and per-bank stock all hang off it.



- **Admins** register the `blood_banks` (facilities), the people, and each bank's `blood_stock`.
- **Members** register once (donor + recipient in one) and create `blood_requests` with an urgency and optional deadline; the app computes a `priority_score`. A Critical Emergency also writes a member-facing `notices` alert.
- **Officers** belong to a bank. The **Emergency Board** ranks open requests by priority; `match_donor.php` runs `find_matching_donors()` ($\text{ABO/Rh score} \times 56\text{-day} \times \text{availability} \times \text{nearest}$) over the members pool; `process_donation()` writes the `donations` row in one transaction that resets the member's clock, **banks the units into the officer's bank**, updates the request, and clears that bank's alert.
- **Members** use the **Blood Bank Locator**: it ranks banks nearest-first and draws an OSRM driving route to the chosen one.

Nine tables, and who touches them

TABLE	HOLDS	WRITTEN BY	READ BY
login	Admin accounts	seed	admin login
blood_banks	Facilities (name, address, geocoded lat/lng, contact)	admin (manage banks)	locator, stock, officers, reports
hospital_workers	Officers (+ blood_bank_id, position)	admin (add officer → picks a bank)	officer login, matching, reports
members	Unified accounts — donor + recipient (blood group, 56-day clock, lat/lng, availability , member_code)	self-registration, admin add member	login, matching, Find Donors, requests, map, reports
blood_requests	Requests (+ urgency, severity , required_within_hours , priority_score , status, lat/lng); recipient_id → the member who asked	member (request_blood)	queue, Emergency Board, match, history
donations	Donation ledger; donor_id / recipient_id both → members, + blood_bank_id	officer (process_donation)	histories, reports
blood_stock	Units per bank per blood type	admin, process_donation	stock page, dashboard, locator, reports
stock_alerts	Low-stock alerts per bank per type	stock page, process_donation	dashboard, stock page
notices	Emergency alerts	officer/admin, auto (critical request)	member emergency alerts

This matches the brief's ERD (Figure 5.3): the design centres on `blood_banks`, with `blood_stock` and `stock_alerts` scoped per bank and `hospital_workers` belonging to a bank. The former separate `donors` and `recipients` tables are now one `members` table — `blood_requests.recipient_id` and

`donations.donor_id` / `recipient_id` keep their names (they describe the role in each transaction) but reference members.

04 / Role walkthroughs

What each role does, end to end

ADMINISTRATOR runs the network

Signs in at `login.php` (`admin` / `admin`) → `dashboard.php` .

- **Blood Banks:** add / edit / delete facilities — each address is geocoded onto the map, and a new bank gets an empty stock row per blood type automatically.
- **Members & Officers:** Manage Members (one CRUD for every donor/recipient — add, edit, delete; member codes auto-generate) and Hospital Officers (each assigned to a bank); passwords hashed on create.
- **Blood Stock:** pick a bank, then set units + low-stock threshold per type; dropping below threshold raises that bank's alert, restocking resolves it.
- **Emergency Board & Find Donors:** see every open request ranked by priority with live countdowns, and search the member pool by group, compatibility and availability.
- **Oversight:** the Blood Requests queue (read-only for admin), Emergency Alerts, the Geo-Map/locator, and Reports — with a blood-group chart, a donation-status doughnut and a stock-by-group chart, plus CSV export.

HOSPITAL OFFICER makes the match

Signs in at `user-login.php` (Hospital Officer, email + password) → `officer_dashboard.php` . Each officer belongs to one bank.

- **Emergency Board:** open requests ranked by `priority_score` , each with a live countdown to its deadline. Work the top of the list first.
- **Blood Requests & Find Donors:** the open queue, and a search over the member pool (available-first, best compatibility, nearest).

- **Match & confirm:** `match_donor.php` lists ABO/Rh-compatible, 56-day-eligible members with a **Match score** and an **Availability** flag, ranked available-first then by score then nearest. Confirming runs `process_donation()`, which banks the units into *this officer's bank*.
- **Emergency Alerts & Geo-Map.**

MEMBER is both supply and demand

Registers at `register.php` (blood group + geocoded address), signs in with **email + password** → `member_dashboard.php`. One account covers both giving and needing blood.

- **My Availability:** set status (available or not), a schedule (anytime, weekends, evenings, emergencies...), and a contact preference. Available members surface first when an officer searches.
- **Donate Blood:** shows eligibility (inside/outside the 56-day window). Officers record the actual donation on match.
- **Request Blood:** patient, group, units, hospital, urgency, and optionally *needed within (hours)*. The app derives a severity and priority score; Critical Emergency auto-posts an alert; one pending request per group is enforced; the hospital is geocoded.
- **Request History** (status + donations received), **Emergency Alerts**, and the **Blood Bank Locator** (Nearby Banks).

05 / The enhancements

Four features that lift it above a basic register

These are the additions that make BloodLink an *enhanced* system — each ties back to the spine in Section 02.

01 • AVAILABILITY REGISTRY

Members say when they can give

Each member keeps an `is_available` flag, a schedule and a contact preference, edited from their own dashboard. The admin/officer **Find Donors** page searches the pool by group and mode (compatible / exact group), optionally filtered to available members and ranked from a chosen bank.

02 • EMERGENCY WORKFLOW

Requests are triaged, not just queued

A request carries a severity and an optional deadline. The **Emergency Board** orders open requests by `priority_score` and shows a live countdown per request, so the most urgent, rarest, soonest-needed case is always at the top.

03 • COMPATIBILITY-AWARE MATCHING

The system scores every donor

Matching goes beyond "same group": a compatibility score rewards exact matches and universal O-, blood-type rarity breaks ties, and available members sort first. The officer sees the score and availability right in the match table.

04 • ANALYTICS

At-a-glance charts

The admin dashboard charts current stock by blood group; Reports adds a member blood-group distribution and a donation-status doughnut (Completed / Pending / Cancelled), alongside the existing KPIs and CSV export.

HOW THE PRIORITY SCORE IS COMPUTED

```
priority = severity (critical 60 · severe 40 · moderate 20 · low 5)
          + rarity × 2 (AB- rarest 10 ... 0+ most common 3)
          + time pressure (≤3h 20 · ≤12h 12 · ≤48h 6 · else 2)
          → capped at 100
```

A Critical, AB-, needed-within-2-hours request therefore scores $60 + 20 + 20 = 100$ and leads the board. The compatibility score is 0 if incompatible, else 100 (exact) / 90 (donor O-) / 80 (same ABO) / 70 (otherwise compatible).

06 / Blood Bank Locator

The interactive map (brief Fig 5.5)

The locator answers "which registered bank is nearest, and how do I get there?" — proximity only; stock is managed separately.

1

Capture location

"Use my location" (browser GPS) or "Use Kaduna city centre". A blue marker shows where you are.

2

Rank the banks

Each bank's distance is computed with the Haversine formula and the list is sorted nearest-first, with a colored distance badge (green ≤ 5 km, amber 5–15, red >15) and dashed connectors to the three nearest.

3

Route to one

Click a bank's "Directions": OSRM returns a driving route drawn as a polyline **colored by distance band**

, and the route box shows

distance by road + estimated drive time

.

Built entirely on free, open services — Leaflet + OpenStreetMap tiles, Nominatim geocoding, OSRM routing — so there is no API key or billing account, unlike Google Maps.

07 / The end-to-end donation

Follow one unit of blood

1

A member files a Critical O- request

needed within a few hours → a `blood_requests` row (Pending) with a high `priority_score` + an auto `notices` alert.

2

Members see the alert

on their Emergency Alerts page; available O- members are ready in the pool.

3

Officer opens the Emergency Board

the request sits at the top by priority, its countdown running → Match Donor.

4

System proposes donors

only O-, 56-day-eligible members,
available first

, each with a compatibility score and distance to the hospital.

5

Officer confirms

process_donation(): writes `donations` (with the officer's `blood_bank_id`), resets the donating member's clock, adds units to that bank's `blood_stock`, flips the request, resolves that bank's alert.

6

Everyone sees the result

The requester's history shows Fulfilled; the donor is inside a fresh 56-day window; the bank's stock, the dashboard charts and Reports update.

08 / QA runbook

How to test the whole app

Set up once, then run each role's script. The seed is built for this — 5 real banks, members on the 56-day boundary, all eight blood groups, two members marked unavailable, and requests pre-scored for the Emergency Board.

SETUP

```
mysql -u root -proot -e "CREATE DATABASE IF NOT EXISTS donor_app"
mysql -u root -proot donor_app < db/schema.sql
mysql -u root -proot donor_app < db/seed.sql
# app under Local: http://<site>.local/bloodlink/ · re-run schema.sql + seed.sql
```

ROLE	PAGE	USERNAME	PASSWORD
Admin	login.php	admin	admin
Officer	user-login.php	maryam@bloodlink.test	Officer@123
Member	user-login.php	zainab@bloodlink.test	Member@123

A. AUTHENTICATION & ACCESS

- ✓ All 3 logins land on the right dashboard; wrong passwords rejected. A member logs in with email (or member ID) + password.
- ✓ Logged out, a protected page redirects (302); a member can't reach an admin/officer URL.
- ✓ Logout works from the topbar avatar dropdown and the sidebar, every role.

B. ADMINISTRATOR

- ✓ **Blood Banks:** add a bank (address geocoded), edit it, delete it; a new bank shows 8 stock rows.
- ✓ **Members:** add a member (blank Member ID auto-generates; blood type shows), edit them, delete them; the new member can then log in by email.
- ✓ Add an officer (assigned to a bank); the new officer can log in.
- ✓ **Blood Stock:** choose a bank; drop A+ below threshold → that bank's alert appears + dashboard count rises; save again → no duplicate; restock → resolves.
- ✓ **Charts:** dashboard shows the stock-by-group bar; Reports shows the blood-group bar + status doughnut; Total Donations equals the donations count; CSV export downloads.
- ✓ **Find Donors:** search O- (compatible) with "available only" → available O- members listed with a match score.

C. MEMBER → EMERGENCY BOARD → OFFICER (THE SPINE)

- 1 Member files O-, Critical Emergency, needed within 3 hours**
 - ✓ Shows Pending in their Request History; a matching alert exists; the request gets a high priority score; duplicate/invalid submissions refused.
- 2 Officer opens the Emergency Board**
 - ✓ The new request ranks at/near the top by score; its countdown ticks down; lower-priority requests sit below.

3

Officer matches

- ✓ Only O- members; available-first, each with a match score and distance; day-30 member excluded, day-56 included.

4

Officer confirms

- ✓ Request → Fulfilled/Approved; the officer's bank's O- stock changes; the donation row carries that bank_id; re-confirming the same donor is refused.

D. MEMBER AVAILABILITY

- ✓ On the member dashboard, toggle availability off, change schedule + contact preference, Update → confirmation shows and the values persist on reload.
-
- ✓ An unavailable member drops below available ones in Find Donors (and off the list with "available only" ticked).

E. BLOOD BANK LOCATOR

- ✓ "Use Kaduna city centre" ranks banks nearest-first with colored distance badges.
-
- ✓ Click "Directions" → a colored road route draws and the box shows distance + drive time.
-
- ✓ OpenStreetMap attribution is visible; banks show units in stock in their popup.

F. CROSS-CUTTING UI/UX

- ✓ No content sits under the fixed topbar/sidebar; tables scroll rather than stretch.
-
- ✓ Consistent chrome across all logged-in pages, all three roles.
-
- ✓ Accented text renders correctly (e.g. "Officer – Hospital"); phone width collapses the sidebar to a hamburger.
-
- ✓ A tampered/expired form token is rejected (CSRF).

KNOWN LIMITATIONS (EXPECTED)

- **Nominatim rate limit (1/sec):** registering banks/members rapidly may store NULL coordinates — they won't appear on the map until re-saved.
- **OSRM demo server** is fine for a demo, not production traffic; self-host it for real deployment.
- **Admins can't confirm donations** by design — the officer is the clinical actor, and the donation's bank comes from the officer's bank.
- **Rotate the old Google Maps key** in the Google console — it's out of the code but in earlier git history.

BloodLink Operations & QA Manual (v5) · roles: admin / officer / member (donor + recipient in one) · spine: blood_requests → priority_score → find_matching_donors → process_donation → donations (banked per facility) · enhancements: availability · emergency board · scored matching · analytics · locator on OpenStreetMap + Nominatim + OSRM · 9 tables centred on blood_banks.